

Corvinus University of Budapest

Very nice! We work on a very similar idea.  
lets talk if you want to do this. :)

# The public transport's impact on the amenities in Budapest

Restructure

1.) Problem

2.) Literature rev.

3.) Questions

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Challenges

↳ network centrality overlaps with distance from the center

↳ endogeneity issues

*Economic Complexity teacher: Balázs Lengyel*

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## Research Problem

Amenities in Budapest are concentrated in the inner city, where the public transport network is most connected. Peripheral neighborhoods, by contrast, face limited variety and choice. The location logic is economic: amenities enter where the reachable customer base is largest and easiest to access, i.e., where stop density and network connectedness are high.

Our premise is that PT capability shapes this reachable market. In outer areas with weak network connectivity, expected revenue is lower and entry risk is higher, so fewer amenities appear. Upgrading PT topology in these neighborhoods should expand attractiveness, reduce entry risk, and make new amenity investment financially viable, increasing both amenity density and variety over time.

## Question

This study investigates the extent to which differences in neighborhood-level public transport capability are associated with differences in the density and mix of amenities across Budapest. The investigation is approached from an economic complexity perspective: cities develop through the accumulation and recombination of capabilities. Public transportation may work as an enabling infrastructure for these capabilities by lowering frictions for workers, consumers, suppliers, and services to meet, thereby shaping where diverse activities co-locate. If PT concentrates reach and reduce time costs unevenly, neighborhoods may diverge in their ability to sustain varied amenities.

Our main question is the extent to which variation in public transport capability across Budapest's neighborhoods accounts for cross-neighborhood variation in the presence of amenities-measured both in terms of intensity and variety.

Public transport capability is an availability  $\times$  importance measure, meaning that neighborhoods with more stops should not automatically score higher unless those stops are well connected in the citywide network. A stop's connectedness reflects its structural role in enabling movement and interchange. For this research, connectedness is captured by a stop connectedness score that prioritizes:

citywide

is this city-specific

might be more prevalent in low-income neighborhoods

there is no PT in USA

citywide

what is the direction of causality  
PT  $\rightleftharpoons$  amenities

~~How do you create the network?~~

- Degree centrality: distinct serving lines, considering overlapping variants as one line,
- Transfer centrality: the stop's role as a transfer node in the stop-to-stop network, giving emphasis to nodes that lie on many shortest paths between other stops,
- Network reach: consider proximity to many other well-served stops; a stop linked to other hubs counts more than an isolated one.

*We have this from GTFS*  
If we have the opportunity and the relevant data to measure, we may extend our measures and the analysis with the dimension of travelling time between two stops.

The neighborhood public transportation capability is then the centrality-weighted sum of these connectedness scores within each neighborhood, normalized by population or area to allow a comparison across differently sized areas.

Amenity supply is measured in complementary terms: amenity density (how many venues), amenity diversity (how many distinct categories), and amenity complexity (rarity-weighted presence of higher-order categories). Our research will investigate whether neighborhoods with higher stop-weighted transport capability systematically have richer amenity environments along these dimensions.

With our main question in mind, we formulated a few sub-questions:

- RQ1: Which measure of stop connectedness (route connectedness, transfer centrality, network reach) aligns most strongly with amenity outcomes?
- RQ2: Does the relationship show stable patterns across core vs periphery, and across the Danube (Buda vs Pest)?
- RQ3: Where do mismatch zones emerge - transportation-strong yet amenity-poor neighborhoods, or vice-versa - and what do these outliers suggest about missing complements beyond the transport system?

Based on these, we formulated our hypotheses too:

- H1: Neighborhoods with higher centrality-weighted transport capability display higher amenity density and a greater category variety.
- H2: The relationship is positive, even when neighborhoods are compared having similar centrality and resident density.

*controlling for such variables is not a new H*

*It's look bad if you only list them*

*embed in theoretical argument and explain expectations*



- H3: Transfer-central stops (high structural importance) drive the strongest association with higher-order, rarer amenities. *WHY?*
- H4: Distinct clusters of transport-strong yet amenity-poor and transport-weak yet amenity-rich neighborhoods are expected to exist, because other local barriers outside transport hold amenities back, or strong local assets make up for weak transport. *I don't get this.*

## Literature review

### Economic complexity and urban capabilities → *How does this relate to your problem?*

Economic complexity treats development as the accumulation and recombination of distributed capabilities. Hidalgo and Hausmann (2009) formalize this idea in a bipartite country-product network and show that places producing a diverse set of less ubiquitous outputs possess richer capability sets that underpin growth (Hidalgo & Hausmann, 2009). In our context, neighborhood amenity portfolios can be read as local capability bundles: rarer, higher-order amenities depend on a denser concentration of complementary resources nearby. *None of Hidalgo's papers are connected, I think.*

At the metropolitan scale, complex activities concentrate disproportionately in large cities. Using U.S. data on patents, scientific papers, employment, and GDP, Balland et al. (2020) show the spatial concentration of productive activities rises with their complexity; biotechnology and semiconductors agglomerate more than apparel or paper (Balland et al., 2020). It implies that when matching and exchange are easy, cities can sustain more complex activity mixes, including a wider range of consumption amenities. *sure? the paper is about output*

Glaeser, Kolko, and Saiz's consumer city paper adds the demand-side mechanism: people value access to a variety of consumption amenities, and such amenities help attract and retain residents and firms (Glaeser et al., 2001). This frames neighborhood amenity density/diversity not merely as by-products of production clusters but as first-order urban assets that shape location choices—assets that public transport can enable by expanding effective reach. *Maybe drop?*

### **Accessibility and land use: why connectivity should map to amenities**

Accessibility—the mass of opportunities reachable from a place—has long explained urban structure. Hansen (1959) showed empirical links between accessibility and development patterns, pioneering location-based accessibility measures (Hansen, 1959). Geurs and van Wee (2004)

\* centrality is correlated with central location  
↳ subcenters might be identified and compared

synthesize the concept and methods, distinguishing location-based indicators that can operate without schedule data—directly relevant to our choice of a topology-only public transport capability index (Geurs & van Wee, 2004). These foundations justify expecting stronger PT connectivity to align with richer amenity environments, even if we deliberately do not include headways or scheduling.

At small scales, centrality in accessibility correlates with activity intensity and retail success. For Seoul, Kang (2016) finds that pedestrian accessibility/centrality is positively associated with retail sales at the street-segment level (Kang, 2016). Sevtsuk shows that pedestrian flows can be predicted from the street network: locations that are more central get more foot traffic, which helps nearby amenities survive (Sevtsuk, 2021). This supports our claim that neighborhoods with higher stop-weighted, connectivity-based public transport capability should have more—and more varied—amenities.

### **Public transport networks as graphs: defining stop importance**

Network science supplies operational tools for stop importance. Betweenness centrality (Freeman, 1977) captures the extent to which a node lies on shortest paths and thus enables transfers between many origin–destination pairs (Freeman, 1977). Applied to spatial urban networks, centrality measures (degree, closeness, betweenness) characterize formation and hierarchy across cities (Crucitti, Latora, & Porta, 2006). Metro/PT studies use topology-only analyses to assess complexity and robustness (Derrible & Kennedy, 2010). Together, these literatures support our stop connectedness score, namely combining route distinctness (lines per stop), transfer centrality (betweenness on a stop-to-stop graph), and network reach (proximity to other well-served stops), and our aggregation of these scores to neighborhood-level PT capability.

Public-transport–specific research extends centrality to vulnerability and robustness. Cats and Jenelius (2014) develop a dynamic, stochastic notion of PT network vulnerability, showing how network properties identify critical nodes and links whose failure disproportionately harms connectivity (Cats & Jenelius, 2014). This supports weighting stops by structural role rather than raw counts, consistent with our availability  $\times$  importance definition of PT capability.

### **Do transit improvements change nearby amenities?**

Quasi-experimental evidence suggests that improving rapid transit access often increases nearby amenity density and multifunctionality, while effects vary by context. Sonnenschein, Arribas-Bel, and colleagues show that subway expansion across European cities raises local amenity density and variety, with especially strong responses in social amenities (restaurants, arts/nightlife), and that existing subcenters benefit most, which is consistent with agglomeration mechanisms (Sonnenschein et al., 2022). While our study is cross-sectional and non-causal, these findings provide empirical priors that access can increase amenity formation where complementary conditions exist.

### **Measuring amenity density, diversity, and complexity**

Scholars state that diversity should be measured with effective numbers (Hill numbers) rather than raw entropies: Jost (2006) shows how to convert entropy and Gini into interpretable “true diversity” counts, yielding a unified family of measures (Jost, 2006; Hill, 1973). We apply this logic to amenity categories: density, diversity, and a rarity-weighted index that emphasizes less ubiquitous amenity types as higher-order capabilities.

### **Amenity complexity and socio-economic mixing: direct Budapest evidence**

Juhász et al. (2023) adapt economic complexity methods to amenity portfolios at the urban-location scale and build an amenity complexity measure from the presence of non-ubiquitous amenity types in Budapest. They couple this with mobility-based evidence on socio-economic mixing, inferring visitor status from the real-estate price of home locations. Two results matter for our study: first, urban centrality is a strong predictor of socio-economic mixing at visited “third places”; second, beyond centrality, both neighborhood complexity and amenity complexity add explanatory power, whereas simple diversity/ubiquity counts do not. Conceptually, they show that rarity-weighted amenity composition captures a form of urban sophistication that relates to inclusive mixing. This directly supports our choice to model amenity complexity alongside density and diversity, and to weight rarer categories more heavily.

The paper’s Budapest-specific construction of amenity complexity means taxonomy, data specifics, and spatial scales are directly comparable to our project. This reduces external validity concerns and supports replication with our PT capability covariate.

*Pinter and Lengyel 2025*

*people had to cross barriers (river, highway, etc.) to ~~access~~<sup>visit</sup> complex amenities*



## Data

We collect the locations and types of different amenities across Budapest by scraping Google Maps data. We classify the amenities into several categories and by these categories we calculate an Amenity Complexity Index (ACI) for every neighborhood. We are also planning to use the method of Ripley's K-function to evaluate the distribution of the amenities within the neighborhoods. We expect higher density around hubs where several public transport lines cross each other. However, the ACI is probably higher in those areas that are closer to the city center than the outer neighborhoods.

OSM?  
Google is  
not for  
free

From BKK's static data (GTFS/stop geometry) we will extract the locations of transit stops and the topology of the stop-to-stop network (no timetable). For each neighborhood, we will calculate stop counts and densities (also normalized by population), and derive stop-level centrality metrics (degree, betweenness, and eigenvector centrality) based on graph connections. Our expectation is a greater concentration and variety of amenities around stops with higher centrality.

We have a  
time table  
network  
Night/day

We will create maps of Budapest showing neighborhood ACI values, amenity locations, and the transit network (with high centrality stops highlighted).

across would  
be interesting to  
compare

## Policy suggestion

When a neighborhood's amenity density/diversity is low, raise its public transport capability to increase the number of residents in the given areas; higher accessible demand increases the likelihood that new amenities enter, and existing ones persist. BKK should use topology-based public transport capability vs amenity maps to re-site or consolidate stops to shorten transfers, add missing connective links so distinct lines intersect, and simplify overlapping variants so the node gains more distinct lines. District municipalities may align station-area public-realm works with PT upgrades (continuous sidewalks, safe crossings, wayfinding, short mid-block links), deploy micro-interventions at flagged hubs (sidewalk continuity, bike parking, lighting, wayfinding, market pitches), and fast-track licenses for missing basic services within walking distance of high centrality stops. Economic development actors can rank opportunity areas where stop centrality is high, and amenity diversity is low, direct small grants to first movers and main tenants, and position incubators adjacent to transfer nodes. The visibility of these effects is expected where PT topology



meaningfully expands accessible demand; if amenity response remains weak after an upgrade, investigate non-transport frictions (zoning, rents, frontage rules) separately.

## **Possible extensions of our research**

A natural extension is to test causality whether amenities concentrate more densely around hubs and, if so, to treat hub formation as a base for policy to steer amenity location. Operationalize hubs as stops or stop-clusters in the top percentile bands of route distinctness, transfer betweenness, and intermodality (e.g. top 10% on any two metrics), and run sensitivity checks to alternative thresholds. If amenity density and complexity are statistically higher within some radius of these nodes, the result supports creating or upgrading hubs in under-served areas to increase attractiveness and lower entry risk for new amenities.

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